

Boolean Algebra

In 1854, George Boole developed an algebraic system now called Boolean algebra.

Boolean algebra is a system of mathematical logic.

It is an algebraic system consisting of the set of elements (0, 1), two binary operators called OR, AND, and one unary operator NOT.

It is the basic mathematical tool in the analysis and synthesis of switching/digital circuits. It is a way to express logic functions algebraically.

AND Law

Law1: $A \cdot 0 = 0$ (Null law)

Law2: $A \cdot 1 = A$ (Identity law)

Law3: $A \cdot A = A$ (Idempotence law)

OR Law

Law1: $A + 0 = A$

Law2: $A + 1 = 1$

Law3: $A + A = A$ (Idempotence law)

Boolean Algebra

Boolean algebra is a useful tool for simplifying digital circuits. • Why do it? Simpler can mean **cheaper, smaller, faster**.

- Example: Simplify $F = x'yz + x'yz' + xz$.

$$F = x'yz + x'yz' + xz$$

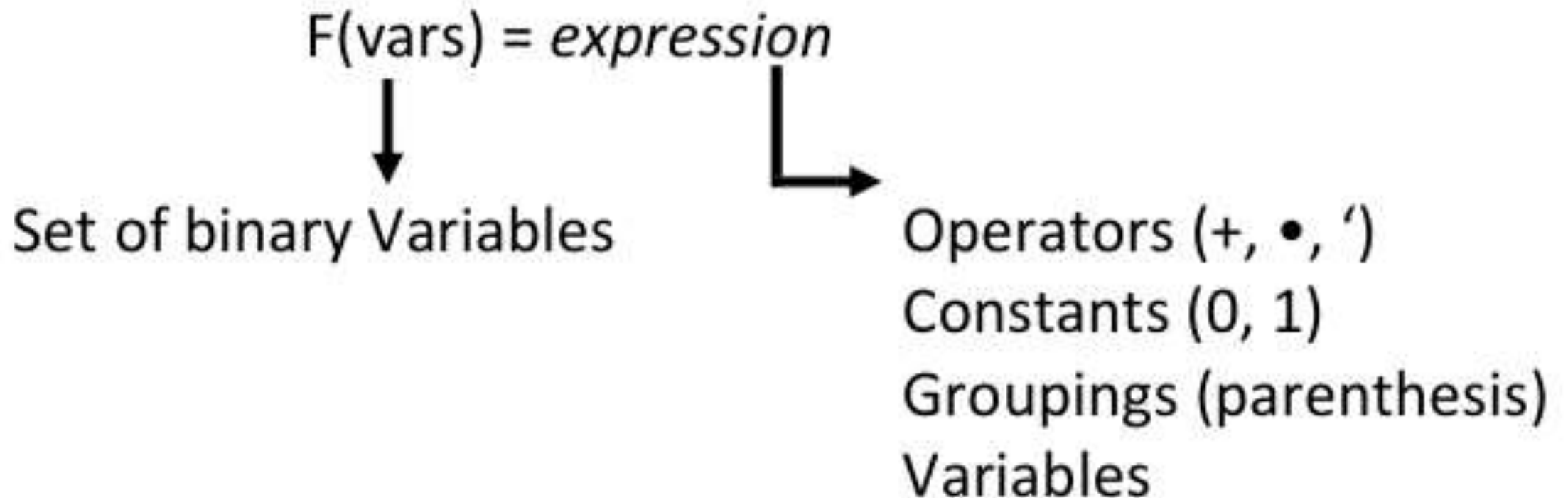
$$= x'y(z+z') + xz$$

$$= x'y \cdot 1 + xz$$

$$= x'y + xz$$

Boolean Function

A Boolean function described by an algebraic expression consists of binary variables, the constants 0 and 1, and the logic operation symbols. For a given value of the binary variables, the function can be equal to either 1 or 0.

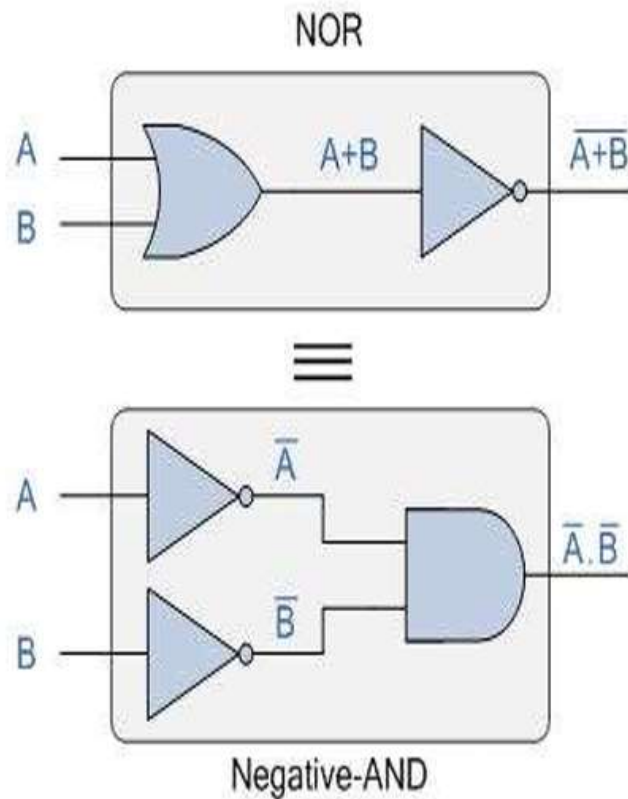


De- Morgan's First Theorem



- It states that the complement of the sum of two or more variables is equal to the product of the complement of individual variables.
- Mathematically,
$$(A+B)' = A' \cdot B'$$

De- Morgan's First Theorem



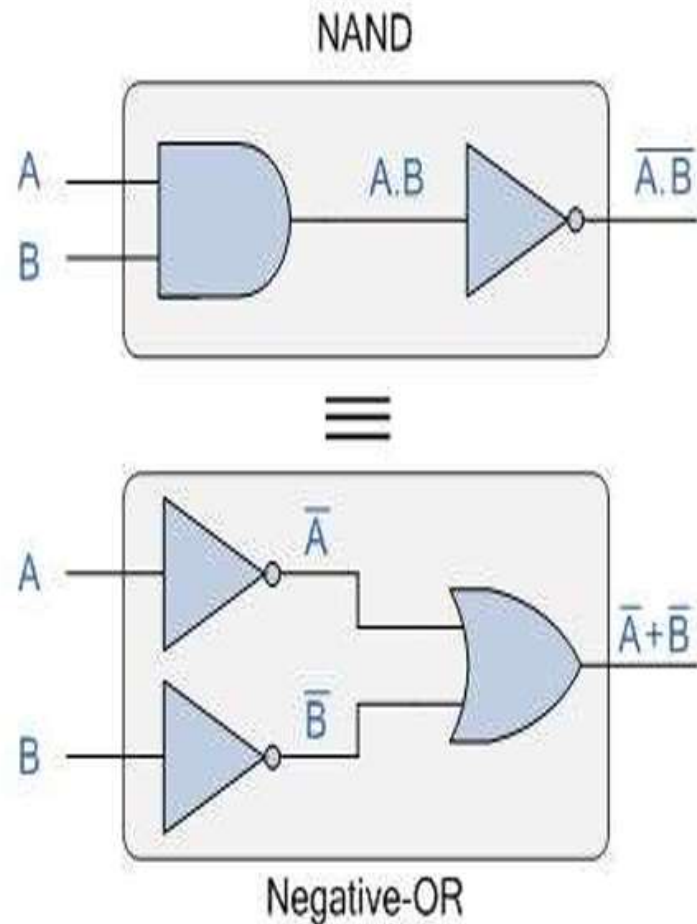
Inputs		Truth Table Outputs For Each Term				
B	A	$A+B$	$\overline{A+B}$	$\bar{A}$	$\bar{B}$	$\bar{A}.\bar{B}$
0	0	0	1	1	1	1
0	1	1	0	0	1	0
1	0	1	0	1	0	0
1	1	1	0	0	0	0

De- Morgan's Second Theorem



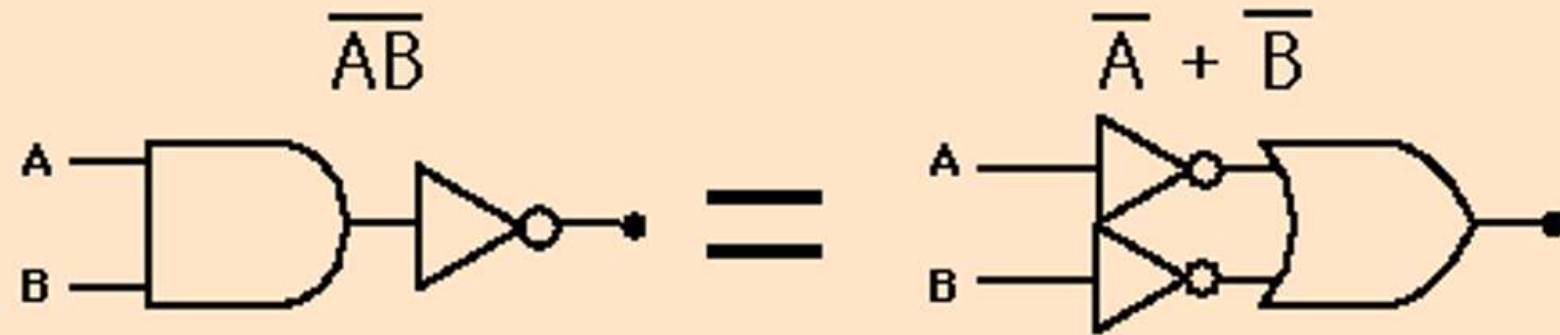
- It states that the complement of the product of two or more variables is equal to the sum of the complements of the individual variables.
- Mathematically,
$$(A.B)' = A' + B'$$

De- Morgan's Second Theorem

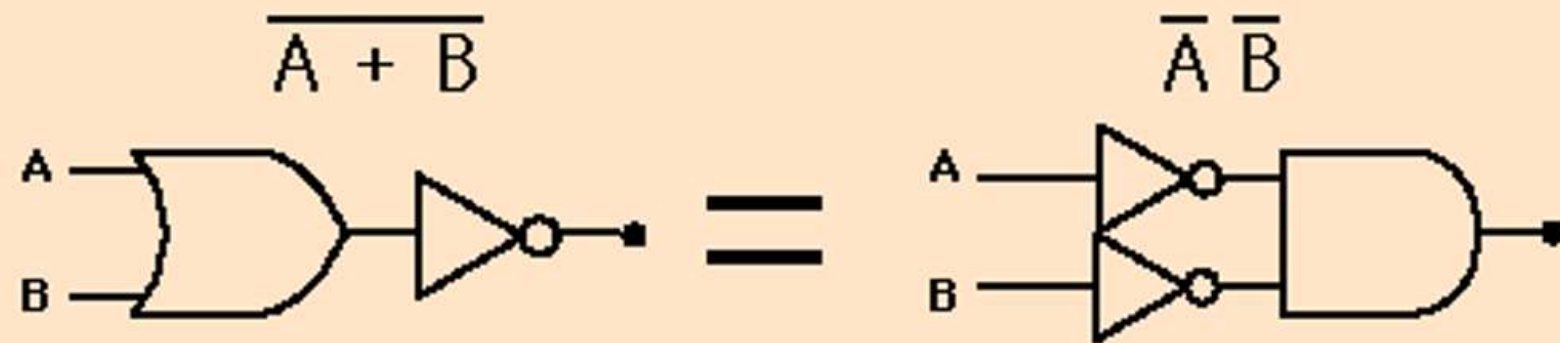


Inputs		Truth Table Outputs For Each Term				
B	A	$A.B$	$\overline{A.B}$	$\overline{A}$	$\overline{B}$	$\overline{A} + \overline{B}$
0	0	0	1	1	1	1
0	1	0	1	0	1	1
1	0	0	1	1	0	1
1	1	1	0	0	0	0

De- Morgan's law



A NAND gate is equivalent to an inversion followed by an OR



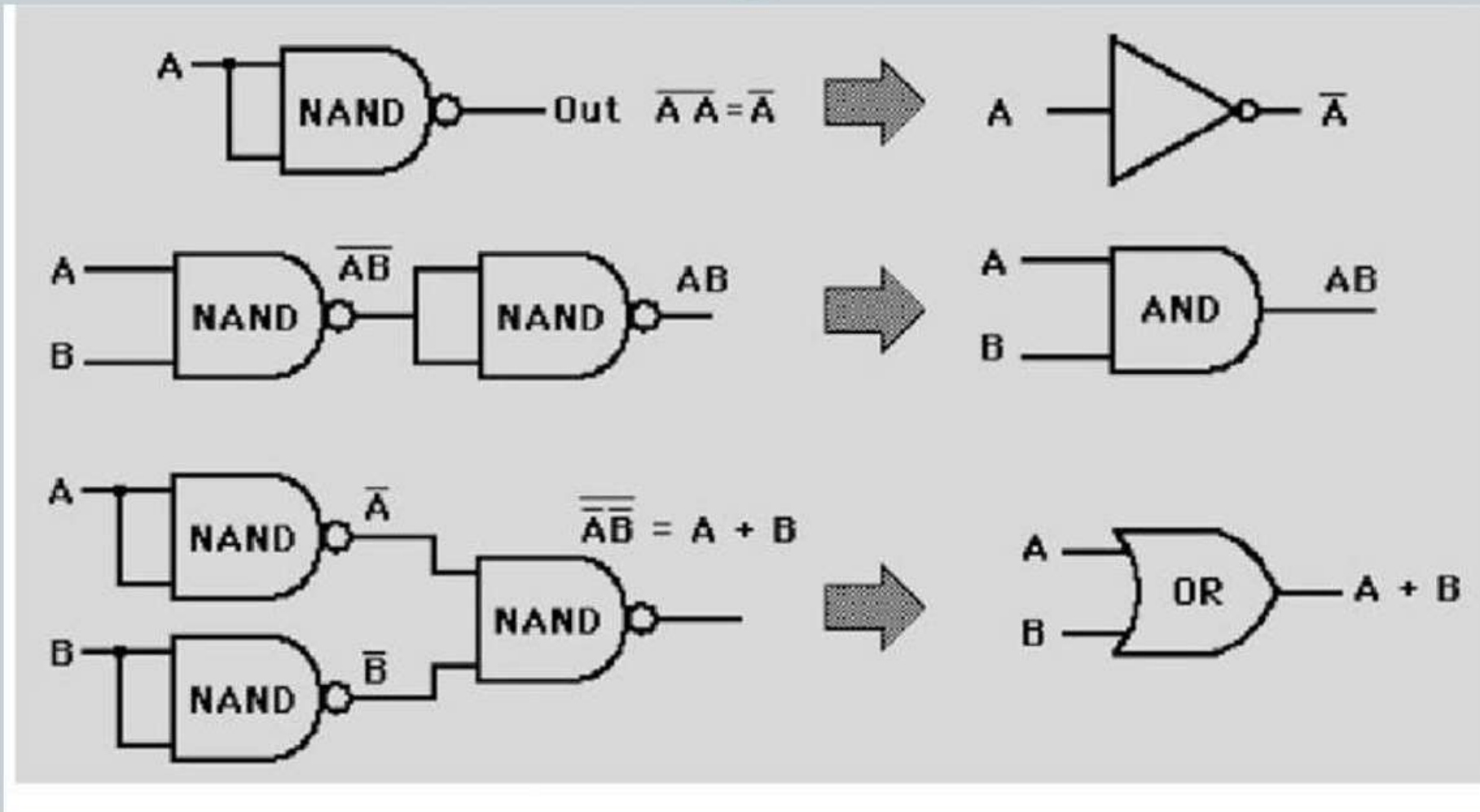
A NOR gate is equivalent to an inversion followed by an AND

Implementation of Universal gates



- Both NAND gate and NOR gate are known as universal gates.
- We can form any other gates(OR, NOT, XOR , XNOR) using these two gates.

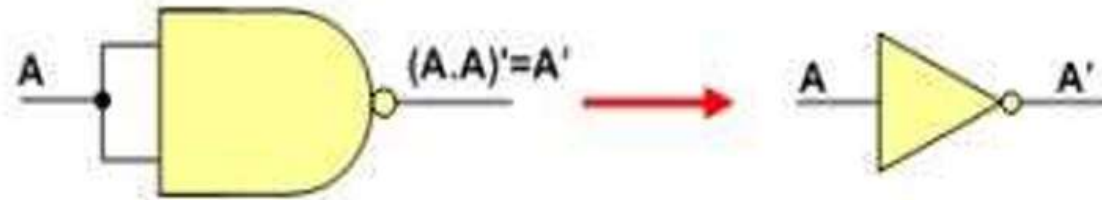
Implementation of Universal gates



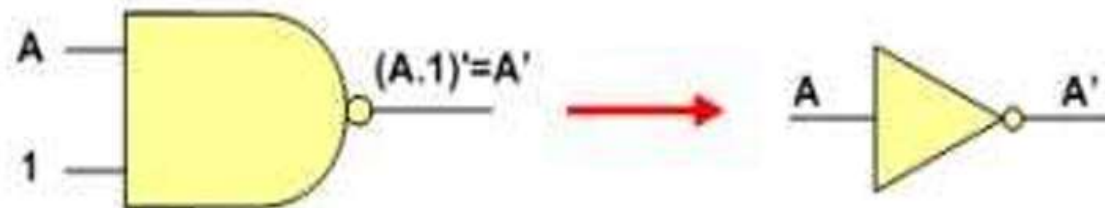
NAND as a universal gate

- Implementing an Inverter Using only NAND Gate.

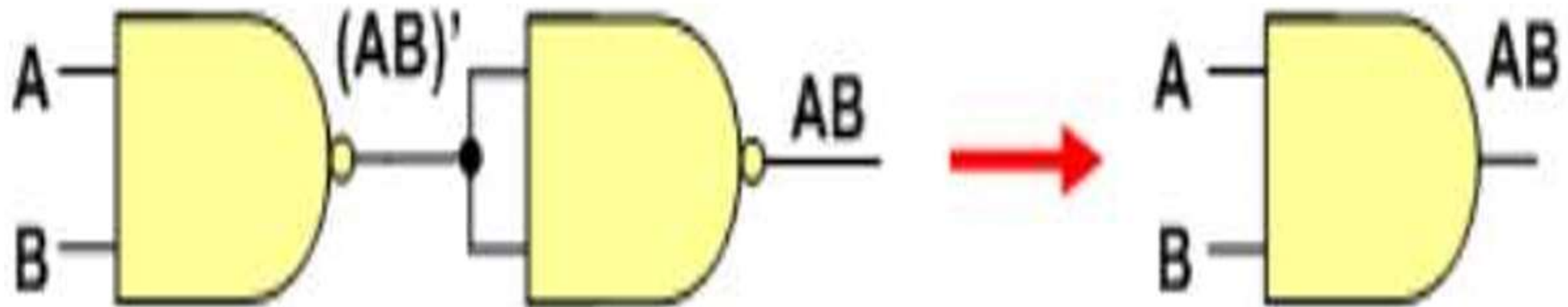
1. All NAND input pins connect to the input signal **A** gives an output **A'**.



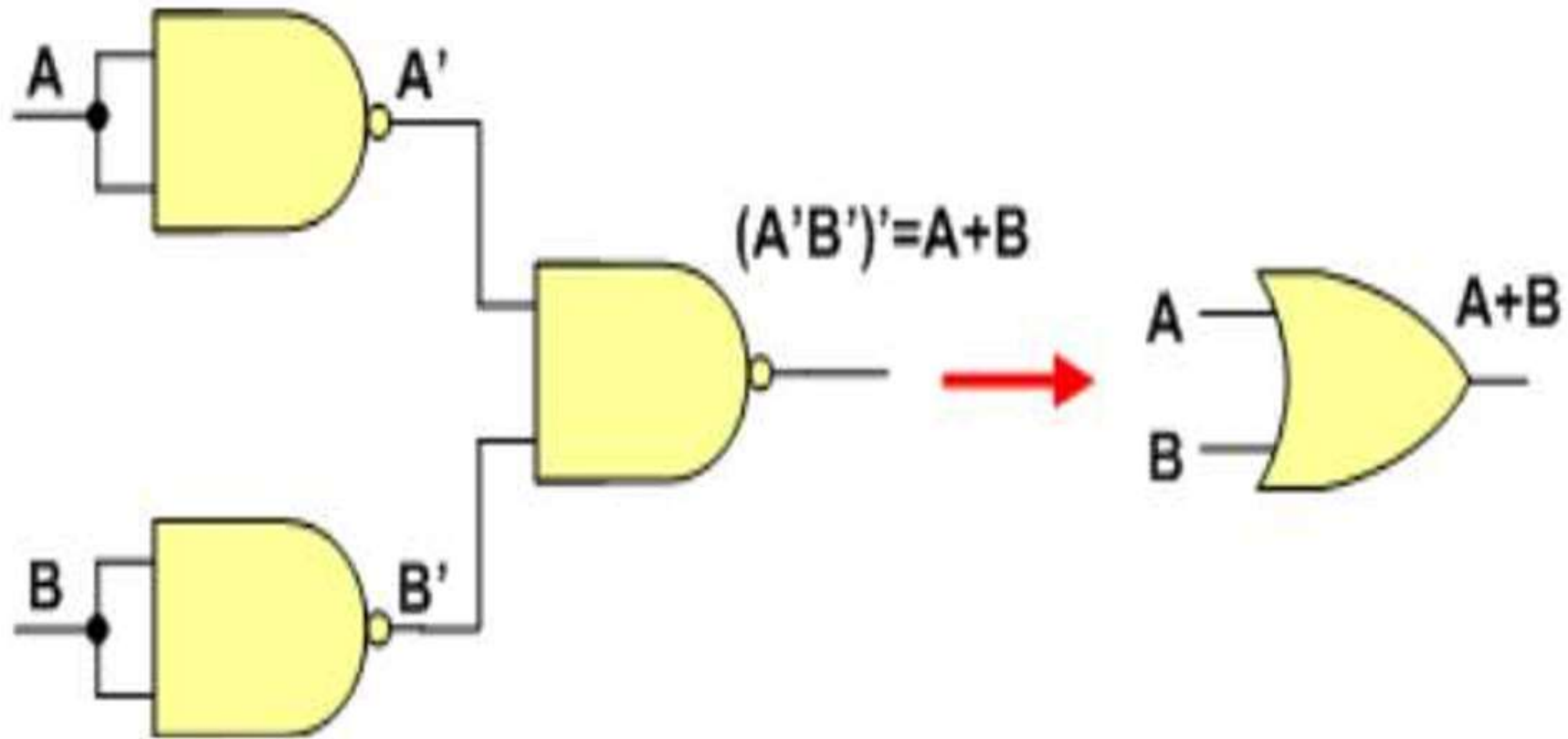
2. One NAND input pin is connected to the input signal **A** while all other input pins are connected to logic **1**. The output will be **A'**.



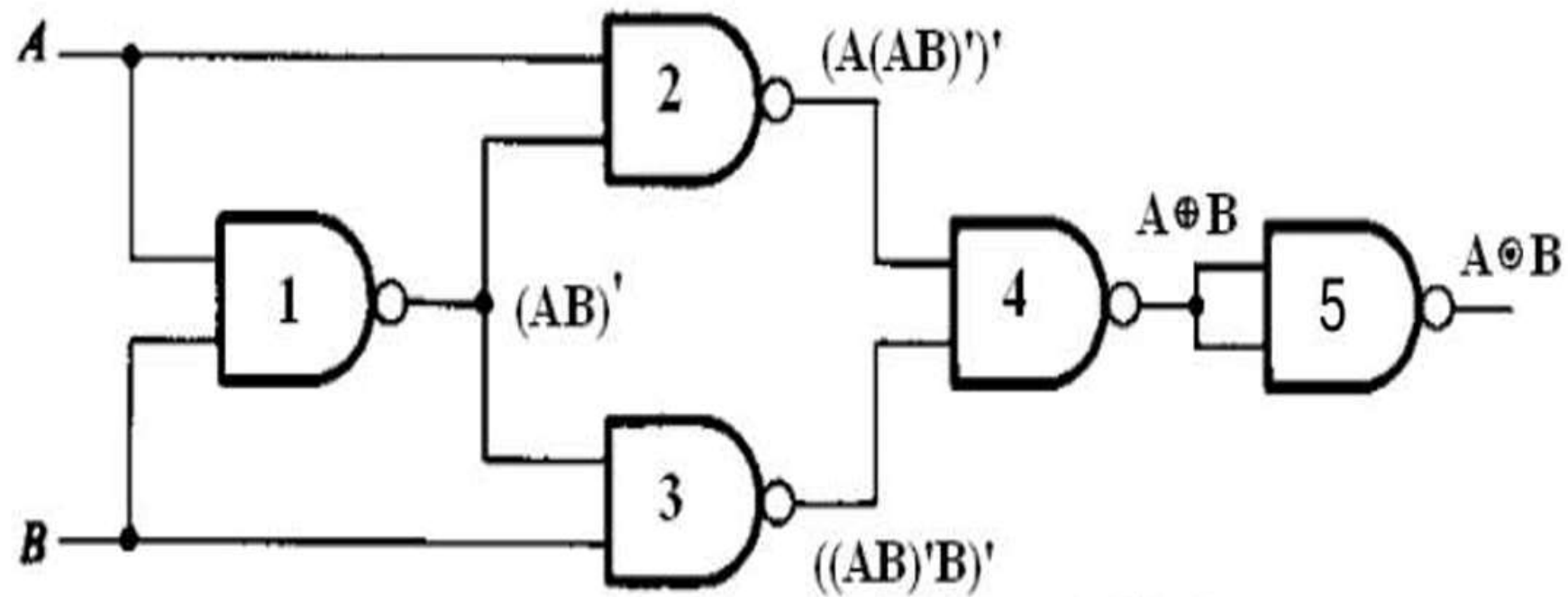
Implementing AND Using only NAND Gates



Implementing OR Using only NAND Gates

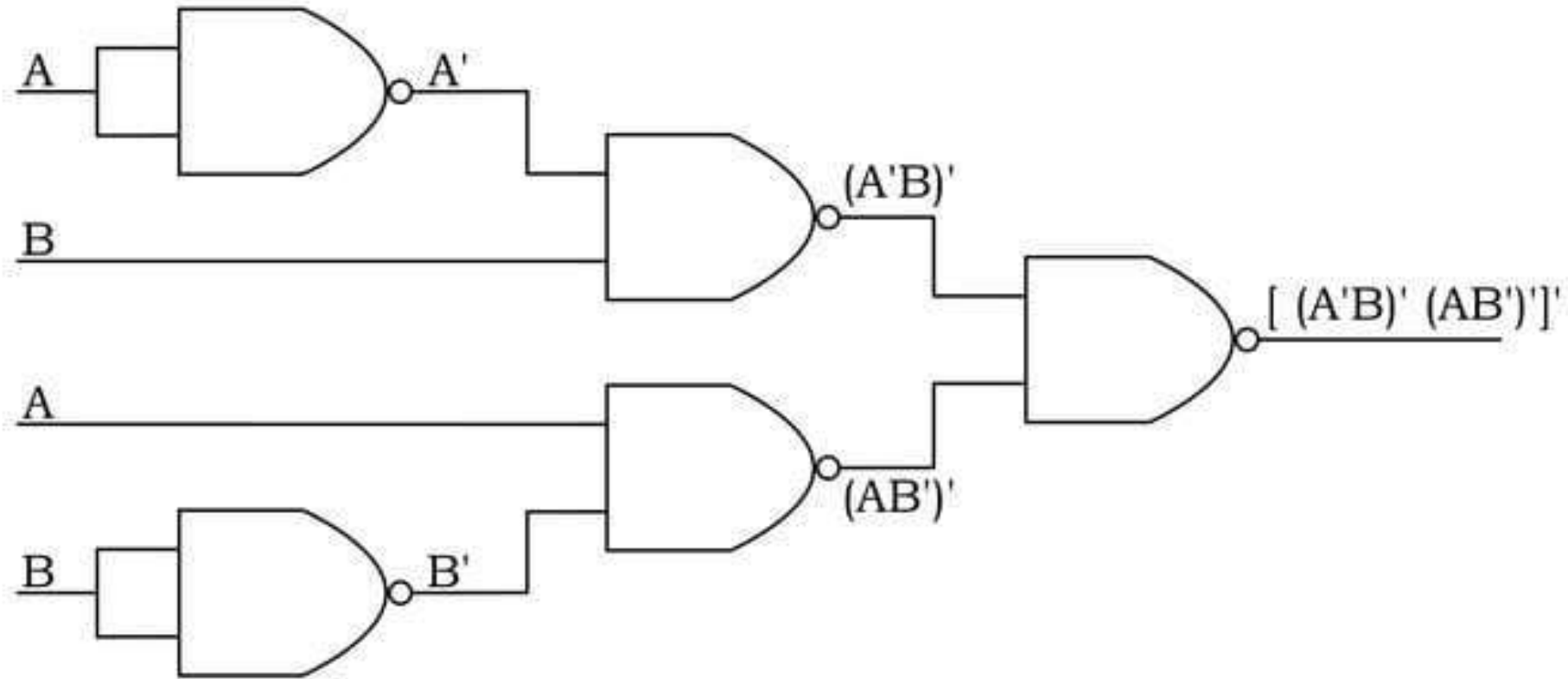


NAND Gates as EX-NOR Gate



X-NOR

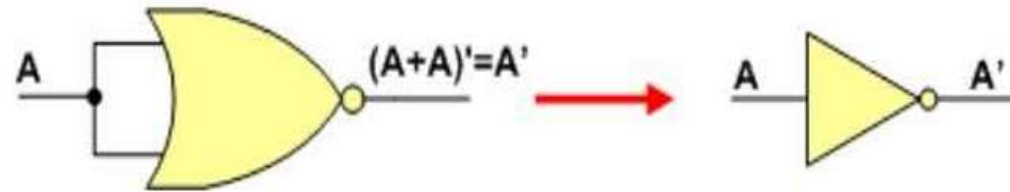
XOR Gate using NAND Gate



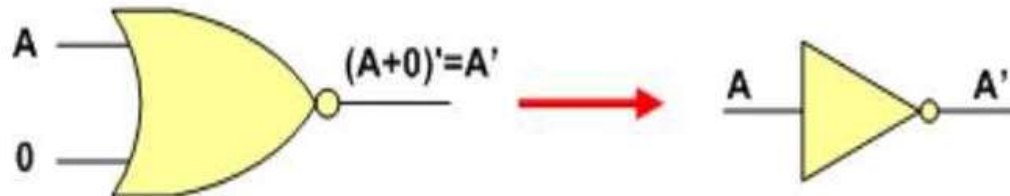
NOR gate as a universal gate

- Implementing NOT gate Using only NOR Gate

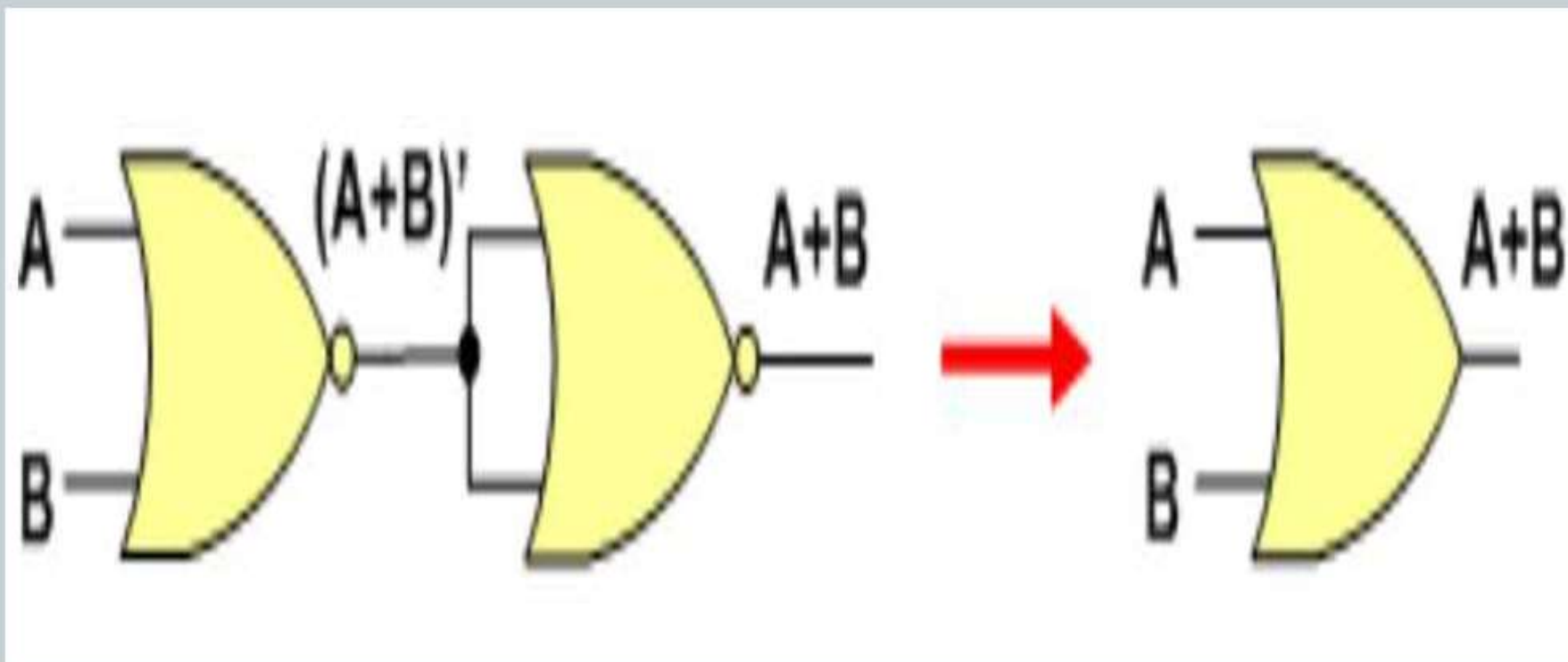
1. All NOR input pins connect to the input signal **A** gives an output **A'**.



2. One NOR input pin is connected to the input signal **A** while all other input pins are connected to logic **0**. The output will be **A'**.



Implementing OR Using only NOR Gates



Implementing AND Using only NOR Gates

